

DIRECTORY

Science has become both increasingly diverse and increasingly connected. Researchers across the globe collaborate to increase human knowledge in every field and seek to bring their different disciplines closer together in pursuit of bigger, broader, more useful theories.

FRED HOYLE

1915–2001

British astrophysicist Fred Hoyle worked on the origin of elements in stars. He postulated that most chemical elements could be created step-by-step by nuclear reactions within large stars. Hoyle was also a proponent of the steady state theory. This states that as the Universe expands, its average density is kept constant as new matter is continuously created. Hoyle coined the phrase “Big Bang” for the main rival theory during a popular radio talk but used the term disparagingly. The steady state theory was largely discredited in the 1960s. In later life, Hoyle took particular interest in the presence of organic molecules in comets, which he believed had brought life to Earth.

IRENE UCHIDA

1917–2013

A Canadian medical geneticist, Irene Uchida instigated work on how genes and chromosomes affect health. She investigated chromosome differences in various genetic conditions and congenital abnormalities, including congenital heart disease and Down syndrome, in which people have an

extra chromosome (47 rather than 46) in their cells. She also examined how X-rays affect chromosomes.

KATHERINE JOHNSON

1918–

Award-winning American mathematician Katherine Johnson played a key role in calculating the trajectory for many space missions, including those of Alan Shepard, the first American in space. She began work at the National Advisory Committee for Aeronautics (NACA), a forerunner of NASA, in 1953, manually carrying out calculations for the engineers there. She helped to plan the launch site and flight path for the Apollo 11 Moon mission in 1969, and later worked on the US space shuttle program. She retired from NASA in 1986.

YVONNE BRILL

1924–2013

Born near Winnipeg, Canada, Yvonne Brill started work with the Douglas Aircraft company in 1945. She went on to invent a propulsion system that keeps unmanned spacecraft in stationary orbit. She also worked on the thrusters for weather satellites, rocket designs used in US Moon missions, and the

Mars observer. While working for NASA from 1981 to 1983, she worked on a rocket engine for the space shuttle.

ABDUS SALAM

1926–1996

Born in Pakistan, Abdus Salam studied mathematics and physics at the University of Cambridge, UK. He proposed the electroweak theory, which explains how the weak interaction between subatomic particles works and how it could be unified with the electromagnetic force. His theory postulated the existence of messenger particles for the weak interaction, with different messengers being exchanged in different instances of the interaction. His theories were proved correct at CERN (the European Organization for Nuclear Research) in 1973, and he was jointly awarded the Nobel Prize in Physics with Sheldon Lee Glashow and Steven Weinberg in 1979.

VERA RUBIN

1928–2016

During the 1970s, Vera Rubin discovered that stars orbiting in the outer part of a spiral galaxy travel as fast as those close to the center. This was contrary to Newton’s laws of gravity and implied that each galaxy had a halo of invisible matter whose gravitational force affected the outer stars. In 1970, Rubin showed this to be true of the Andromeda Galaxy; by 1985, she had examined 60 spiral galaxies and